

VTE — Wells Score & D-dimer Pretest Probability

THE WELLS CUSTOMS BOOTHS & D-DIMER SENTRY — VTE PRETEST PROBABILITY

NEVER interpret D-dimer or imaging without pretest probability first

1 THE WELLS-DVT CUSTOMS BOOTH

≤0 LOW · 1-2 MOD · ≥3 HIGH

- 1 Active cancer (+1)
- 2 Paralysis, paresis, or recent cast (+1)
- 3 Bedridden >3d OR major surgery <12wk (+1)
- 4 Tenderness along deep veins (+1)
- 5 Entire leg swelling (+1)
- 6 Unilateral calf swelling >3 cm (+1) ☆
- 7 Pitting edema (+1)
- 8 Collateral superficial nonvaricose veins (+1)
- 9 Alternative dx as likely (-2) — **KEY TRAP**

2 THE WELLS-PE CUSTOMS BOOTH

> 4 = HIGH probability

★ **BIG TWO**

- 10 Signs/symptoms of DVT (+3) ☆
- 11 No alt dx more likely than PE (+3) ☆
- 12 HR >100 (+1.5)
- 13 Immobilization >3d OR surgery <4wk (+1.5)
- 14 Prior PE or DVT (+1.5)
- 15 Hemoptysis (+1)
- 16 Cancer (+1)

3 THE D-DIMER SENTRY

- 17 D-DIMER — SENSITIVE >95% (ELISA)
- 18 Rules OUT VTE if NEGATIVE + low/mod
- 19 False (+): pregnancy, sepsis, cancer, cancer, post-op, MI
- 20 NOT USEFUL in hospitalized pts
- 21 AGE-ADJUSTED: >50yo, suspected PE → cutoff = age × 10 ng/mL



4 THE HIGH-PROB EXPRESS GATE



22 HIGH PROB → SKIP D-DIMER → STRAIGHT TO IMAGING (CTPA)

5 THE DVT MIMIC LUGGAGE PILE

- 23 Baker's cyst
- 24 Muscle strain / injury
- 25 Cellulitis
- 26 Chronic venous insufficiency / PTS

6 THE PE MIMIC SHIPS

- 27 Pneumonia / Asthma / COPD
- 28 CHF / Pericarditis
- 29 Rib fx / Pleurisy
- 30 ST change

Wells score FIRST. Low/Mod → D-dimer (age-adjusted >50). High → CTPA directly. D-dimer rules OUT only.

1. Wells DVT booth

WHAT YOU SEE

Customs booth labeled DVT Wells score

WHAT IT ENCODES

The Wells DVT score stratifies pretest probability; the traditional 3-tier read is ≤ 0 low, 1-2 moderate, ≥ 3 high. The dichotomized version is DVT 'unlikely' (≤ 1) vs 'likely' (≥ 2). Low/moderate (unlikely) $\rightarrow$ check D-dimer; high (likely) $\rightarrow$ compression ultrasound. Always assign pretest probability BEFORE ordering any test.

BOARD TRAP

WRONG: 'order a compression ultrasound first on every leg-pain patient regardless of score.' In a low-probability (unlikely) patient a negative high-sensitivity D-dimer rules out DVT without imaging — leading with ultrasound everywhere wastes resources and misses the algorithm.

2. Active cancer (+1)

WHAT YOU SEE

List item active cancer +1 point

WHAT IT ENCODES

Active cancer (treatment within 6 months or palliative) adds +1 to the Wells DVT score — malignancy is prothrombotic via tissue factor expression and is itself a strong VTE risk factor. Other +1 items include paralysis/recent cast immobilization and recent bedridden >3 days or major surgery within 12 weeks.

BOARD TRAP

WRONG: 'a normal D-dimer rules out DVT in a low-Wells patient who has metastatic cancer.' D-dimer is chronically elevated in active malignancy, so it loses specificity and the negative-D-dimer rule-out is unreliable — these patients generally go to ultrasound.

3. Unilateral calf swelling >3 cm (+1)

WHAT YOU SEE

Swollen calf measured >3 cm

WHAT IT ENCODES

Calf swelling >3 cm versus the asymptomatic leg (measured 10 cm below tibial tuberosity) = +1, as do pitting edema confined to the symptomatic leg (+1), tenderness localized along the deep venous system (+1), entire-leg swelling (+1), and collateral superficial (non-varicose) veins (+1). These objective unilateral findings raise pretest probability.

BOARD TRAP

WRONG: 'bilateral leg swelling makes DVT the top diagnosis.' Bilateral edema points toward systemic causes (heart failure, cirrhosis, nephrotic syndrome) — DVT classically causes UNILATERAL swelling, and the Wells criteria reward asymmetry.

4. Alternative dx as likely (−2)

WHAT YOU SEE

Red key-trap item: alternative diagnosis at least as likely

WHAT IT ENCODES

If an alternative diagnosis is AT LEAST AS LIKELY as DVT, SUBTRACT 2 points — the only negative item in the Wells DVT score and the most commonly forgotten modifier. It encodes clinical gestalt: a plausible cellulitis or Baker's cyst should lower your DVT probability.

BOARD TRAP

WRONG: 'tally only the positive criteria and skip the −2.' Omitting the −2 falsely inflates the score and can push an 'unlikely' patient into the 'likely' category, triggering unnecessary imaging — this minus-2 modifier is a classic exam point.

5. Wells PE booth

WHAT YOU SEE

Customs booth labeled PE Wells score

WHAT IT ENCODES

The Wells PE score: dichotomized as ≤ 4 'PE unlikely' vs >4 'PE likely.' PE unlikely $\rightarrow$ high-sensitivity D-dimer (or apply PERC first); PE likely $\rightarrow$ go straight to CT pulmonary angiography. The simplified Wells uses 1 point per item with ≤ 1 = unlikely.

BOARD TRAP

WRONG: 'a PE-likely (Wells >4) patient with a negative D-dimer can be discharged.' At high pretest probability the negative predictive value of D-dimer is inadequate — you must image (CTPA). D-dimer only rules out in low/moderate probability.

6. Clinical signs of DVT (+3)

WHAT YOU SEE

Big-two item: signs/symptoms of DVT +3

WHAT IT ENCODES

Clinical signs/symptoms of DVT (leg swelling, pain on palpation of deep veins) = +3 — one of the 'Big Two' heaviest Wells PE criteria. A coexisting DVT makes concurrent PE far more likely, so this item carries the most weight along with clinician judgment.

BOARD TRAP

WRONG: 'absence of leg symptoms makes PE unlikely.' Many PEs arise from clinically silent proximal DVTs; lack of leg findings does not exclude PE — it simply omits this +3 item while other criteria (tachycardia, gestalt) can still make PE likely.

7. PE most likely diagnosis (+3)

WHAT YOU SEE

Big-two item: no alternative dx more likely than PE +3

WHAT IT ENCODES

'PE is the #1 diagnosis (no alternative more likely)' = +3, the other 'Big Two' criterion. This clinical-gestalt item is heavily weighted and frequently the swing factor that tips a patient over the >4 threshold into 'PE likely' $\rightarrow$ CTPA.

BOARD TRAP

WRONG: 'gestalt doesn't matter, only objective vitals count in Wells.' The single most heavily weighted PE criterion is the subjective 'PE is the leading diagnosis' (+3) — discounting clinician judgment is the error the score deliberately guards against.

8. HR >100 (+1.5)

WHAT YOU SEE

List item heart rate >100 +1.5

WHAT IT ENCODES

Heart rate >100 = +1.5, as do prior DVT/PE (+1.5) and immobilization ≥3 days or surgery within 4 weeks (+1.5). Hemoptysis (+1) and active malignancy (+1) round out the score. Tachycardia is the most common — and most nonspecific — vital sign in PE.

BOARD TRAP

WRONG: 'a normal heart rate argues strongly against PE.' Tachycardia is sensitive-ish but absent in many PEs; a normal HR (and even a normal A-a gradient) cannot exclude PE — use the full pretest-probability pathway.

9. D-dimer sentry

WHAT YOU SEE

Guard labeled D-dimer sensitive >95%

WHAT IT ENCODES

D-dimer is a fibrin degradation product that is highly SENSITIVE (>95% with the ELISA/high-sensitivity assay) but poorly SPECIFIC. Its value is as a rule-OUT test: in a LOW or MODERATE pretest-probability patient, a NEGATIVE high-sensitivity D-dimer effectively excludes VTE and no imaging is needed. Use the high-sensitivity (quantitative ELISA) assay for this purpose.

BOARD TRAP

WRONG: 'a positive D-dimer confirms VTE.' D-dimer rises in countless conditions and can never rule IN VTE — a positive result only mandates further testing (ultrasound or CTPA). It is a rule-out, never a rule-in test.

10. False positives

WHAT YOU SEE

List of false-positive causes: pregnancy, sepsis, cancer, surgery, post-op, MI

WHAT IT ENCODES

D-dimer is falsely elevated in pregnancy, sepsis/infection, active cancer, recent surgery/trauma, recent MI, advanced age, liver disease, and DIC. These conditions wreck its specificity, so a positive value in such patients is uninformative and a negative value may be hard to obtain.

BOARD TRAP

WRONG: 'use a standard D-dimer cutoff to exclude PE in a third-trimester pregnant patient.' Pregnancy physiologically raises D-dimer so the standard cutoff rarely returns negative; pregnancy-specific algorithms (e.g. YEARS) or imaging (lower-extremity US, then V/Q or low-dose CTPA) are used.

11. Not useful in hospitalized patients

WHAT YOU SEE

Note: D-dimer not useful in inpatients

WHAT IT ENCODES

In hospitalized, post-op, or acutely ill inpatients D-dimer is almost universally elevated, so it carries essentially no rule-out value in this population. For these patients move directly to definitive imaging rather than ordering a D-dimer you already know will be positive.

BOARD TRAP

WRONG: 'order a D-dimer to screen a post-op inpatient with new dyspnea.' It will be elevated from surgery regardless of clot status and cannot exclude PE — go straight to CTPA in the symptomatic hospitalized patient.

12. Age-adjusted D-dimer

WHAT YOU SEE

Note: age-adjusted cutoff = age × 10 if >50

WHAT IT ENCODES

For patients >50 years, use an AGE-ADJUSTED cutoff = age × 10 µg/L (FEU) instead of the fixed 500. This raises specificity (fewer needless CTPAs in older patients) while preserving sensitivity, validated in the ADJUST-PE study. Apply only to low/moderate (PE-unlikely) probability.

BOARD TRAP

WRONG: 'apply the age-adjusted cutoff to a high-probability patient to avoid CTPA.' Age adjustment is only valid for non-high pretest probability; a high-probability patient still goes directly to imaging regardless of D-dimer value.

13. High-prob express → skip D-dimer

WHAT YOU SEE

Express gate: high probability bypasses D-dimer to CTPA

WHAT IT ENCODES

High pretest probability (Wells PE 'likely', >4) → BYPASS D-dimer and go straight to CTPA (or V/Q if contrast contraindicated/renal failure/pregnancy). At high probability a negative D-dimer cannot lower post-test probability enough to be safe, so the test adds nothing.

BOARD TRAP

WRONG: 'order a D-dimer in a high-probability patient and send them home if it's negative.' A negative D-dimer in a high-probability patient is a false reassurance — the residual PE risk remains too high; imaging is mandatory.

14. DVT mimics

WHAT YOU SEE

Luggage pile: Baker's cyst, muscle strain, cellulitis, chronic venous insufficiency

WHAT IT ENCODES

DVT mimics that should trigger the Wells '–2 alternative diagnosis': ruptured Baker's cyst, muscle strain/tear/hematoma, cellulitis, and chronic venous insufficiency/post-thrombotic syndrome. PE mimics include pneumonia, asthma/COPD exacerbation, pericarditis, pleurisy, rib fracture, pneumothorax, ACS, and anxiety. Recognizing a strong mimic lowers pretest probability.

BOARD TRAP

WRONG: 'unilateral warm erythematous swollen calf must be DVT, start anticoagulation.' Cellulitis presents identically (warmth, erythema, swelling, tenderness) — confirm with ultrasound before anticoagulating, since treating cellulitis as DVT (or vice versa) is a classic miss.